

Prompt Engineering

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Things to be discussed

- What is a prompt
- Why it's called prompt engineering
- Prompt Examples
 - Role Play
 - Retrieval Augmented Generation [RAG]
 - Meta Prompting
 - One Shot, Few Shot learning
 - Chain of thoughts
 - Multi-Turn Prompting
 - Output Structuring Prompting
 - LLM as Judge

What is a prompt

A prompt is the input or instruction provided to a language model or generative AI system to elicit a response.

It serves as a guide for the AI to generate text, perform tasks, or produce specific outputs.

A prompt can take various forms, such as:

- Questions: What is the capital of France?
- Commands: Write a summary of the given text.
- Context with Instructions: Based on the following data, explain the trends in a concise manner.

Understanding of Tokens

Tiktokenizer

gpt-4o

User

What is 2+2?

×

Assistant

2+2 = 4

×

User

What if it was *?

×

Assistant

2*2 = 4, same as 2+2!

×

Add message

```
<|im_start|>user<|im_sep|>What is 2+2?<|im_end|>
<|im_start|>assistant<|im_sep|>2+2 = 4<|im_end|>
<|im_start|>user<|im_sep|>What if it was *?<|im_end|>
<|im_start|>assistant<|im_sep|>2*2 = 4, same as 2+2!<|im_end|>
```

Token count
50

```
<|im_start|>user<|im_sep|>What is 2+2?<|im_end|><|im_s
tart|>assistant<|im_sep|>2+2 = 4<|im_end|><|im_start|>
user<|im_sep|>What if it was *?<|im_end|>
<|im_start|>assistant<|im_sep|>2*2 = 4, same as 2+2!<|
im_end|>
```

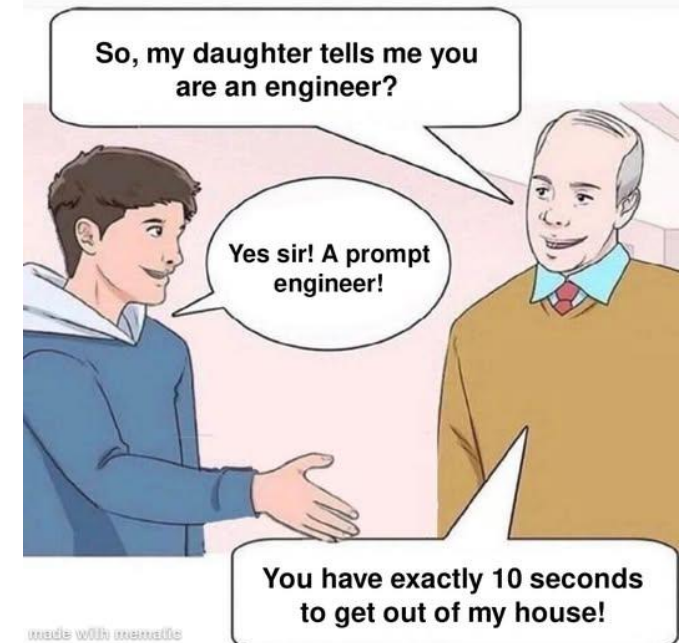
```
200264, 1428, 200266, 4827, 382, 220, 17, 10, 17, 30,
200265, 200264, 173781, 200266, 17, 10, 17, 314, 220,
19, 200265, 200264, 1428, 200266, 4827, 538, 480, 673,
425, 30, 200265, 198, 200264, 173781, 200266, 17, 9, 1
7, 314, 220, 19, 11, 2684, 472, 220, 17, 10, 17, 0, 20
0265
```

What is prompt engineering

The Art of Getting the Best from AI

Prompt engineering is the art of crafting instructions or questions that guide an AI model to generate the desired response or outcome.

The better your prompts, the more accurate and relevant the AI's response will be.



Advantages of prompt Engineering

Increased
Precision and
Relevance

Controlling Bias
and Unintended
Outputs

Improved
Contextual
Understanding

Adapting AI to
Specific Domains
or Use Cases

Enhanced User
Interaction and
Experience

Anatomy of a Good Prompt

What Defines a Good Prompt?

- A good prompt effectively guides the AI model to produce accurate, relevant, and high-quality outputs.
- It should be clear, specific, and tailored to the task at hand. Below are the essential elements of a good prompt:



Prompt Examples

Ask the model to adopt a persona (Role Play)

In prompt engineering, role play refers to instructing the AI model to adopt a specific identity, perspective, or role to generate responses aligned with that persona. By assigning a role, the user can guide the AI to emulate the expertise, tone, or behavior expected from the assumed role.

Importance of Role Play

- Contextual Responses
- Improved Relevance
- Enhanced Creativity
- Problem-Specific Insights

Without Role Play	With Role Play
Prompt: Explain how blockchain works. Output: Blockchain is a decentralized database that uses cryptography to secure transactions and maintain a distributed ledger.	Prompt: You are a blockchain developer. Explain how blockchain works to someone with no technical background. Output: Blockchain is like a digital notebook shared across many computers. Imagine writing something in a notebook, and everyone has an identical copy. If you change something, everyone else's notebook updates too, ensuring the record is accurate and secure.

Prompt Examples

Role Play Example

Without Role Play

Write a song in hindi for the below situation.

Situation

शाम का वक्त है, सूरज धीरे-धीरे पहाड़ियों के पीछे छुप रहा है। आसमान में लाल और सुनहरे रंग बिखरे हुए हैं। ठंडी हवा पहाड़ों से घाटी की ओर बह रही है, मानो कोई धीमी ठंडी सांसे ले रहा हो, पेड़ों की शाखाएँ धीरे धीरे हिल रही हैं, जैसे किसी अधूरे सपने को अलविदा कह रही हों।

उस शांत शाम में एक प्रेमी खड़ा है, उसकी नज़रें दूर उसी चेहरे को खोज रही हैं जिसे वह सबसे ज्यादा चाहता है—अपनी प्रेमिका। कुछ ही पल पहले, सूरज की रोशनी में वह साफ दिख रही थी, लेकिन अब अंधेरा धीरे-धीरे बढ़ रहा है। उसकी आंखें अब उसे देख नहीं पा रही हैं।

उसका दिल भारी हो गया है—वह जानता है, यह उनकी आखिरी मुलाकात है। अब शायद वे कभी नहीं मिलेंगे। जैसे सूरज डूब गया, वैसे ही उनका प्यार भी अंधेरे में खो रहा है। उनके बीच अब बस एक लंबी, गहरी रात रह गई है, जहाँ कोई रास्ता नहीं, कोई उम्मीद नहीं।

हवा अब भी उसकी प्रेमिका की हल्की खुशबू लेकर आ रही है, मानो उसकी आखिरी याद दे रही हो। वह आँखें बंद करता है, उसे चेहरे को अपने मन में हमेशा के लिए संजो लेने के लिए, क्योंकि उसे पता है, जब वह फिर से आँखें खोलेगा, तब वह केवल एक याद बन चुकी होगी—अंधेरे में खोई हुई, हमेशा के लिए दूर।

With Role Play

Imagine you are Javed Akhtar, the legendary poet and lyricist known for your profound and poetic songwriting. Your words paint emotions, your metaphors carry deep meanings, and your lyrics leave a lasting impact on the listener's soul.

Today, you are writing a song on the below situation about separation of two lovers. Let your lyrics flow like a river of emotions, blending nostalgia, hope, and introspection. Use poetic imagery, delicate wordplay, and deep metaphors that touch the listener's heart. The song should feel timeless, just like your most iconic works.

Situation

शाम का वक्त है, सूरज धीरे-धीरे पहाड़ियों के पीछे छुप रहा है। आसमान में लाल और सुनहरे रंग बिखरे हुए हैं। ठंडी हवा पहाड़ों से घाटी की ओर बह रही है, मानो कोई धीमी ठंडी सांसे ले रहा हो, पेड़ों की शाखाएँ धीरे धीरे हिल रही हैं, जैसे किसी अधूरे सपने को अलविदा कह रही हों।

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Example of Role Play

Finance:

You are a Financial Analyst presenting a quarterly performance report to the CFO. The company has missed its revenue targets due to unexpected supply chain delays. Explain the variance, suggest corrective measures, and defend your proposed budget for the next quarter.

HR:

You are an HR Business Partner conducting a conflict resolution meeting between two team members whose disagreements are affecting team morale. Mediate the discussion, uncover the root cause, and propose a mutually acceptable resolution.

Marketing:

You are a Marketing Manager pitching a new digital campaign strategy to the executive team. The goal is to increase brand awareness among Gen Z consumers. Justify the choice of platforms, creative approach, and KPIs.

Supply Chain:

You are a Supply Chain Manager handling a disruption in a key supplier's delivery. Your task is to communicate with operations, logistics, and the vendor to find a contingency plan that minimizes production delays.

Admin:

You are an Administrative Officer briefing a new employee on office policies, resource usage (e.g., meeting rooms, IT equipment), and emergency protocols. Ensure the employee feels welcomed and informed.

Task

Task tells the AI exactly what to do.

Example

Weak:

Tell me about AI customer support solutions.

Strong:

Compare the advantages and disadvantages of building versus purchasing an AI customer support solution.

Impact

Specific tasks produce actionable results.

Context

Context provides background information required for making informed decisions.

More relevant context generally produces better results.

Example

Without Context:

"Suggest a growth strategy."

With Context:

"Our retail bank has experienced a 15% decline in customer acquisition over the past year. Two competitors recently launched AI-powered advisory products. The marketing budget will remain unchanged."

Why it Matters

Context helps the AI understand:

Industry, Business challenges, Available resources, Constraints, Goals

Constraint

Constraints define the boundaries.

Example

- *Budget is limited to \$300,000.*
- *The solution must be implemented within 6 months.*
- *Customer data must remain compliant with regulatory requirements.*

Impact

Without constraints, the AI might recommend unrealistic solutions.

Format

Output format tells AI how to present the answer.

Example

Present the findings in a table with:

- Option
- Cost
- Benefits
- Risks
- Time to Implement
- Recommendation

Impact

Leadership receives a decision-ready output.

Format

Output Structuring Prompting

Output Structuring refers to designing prompts or processes in a way that ensures the generated output follows a specific, organized format. This technique is particularly useful when working with AI systems to extract, summarize, or transform data into structured formats like tables, JSON, or lists for easier interpretation and usage.

You are an expert system designed to extract education information from resumes. Analyze the provided resume text and extract education details. Organize the extracted data into a table with the following columns: Qualification, Stream, CGPA, and Passing Year. If any column is missing or the information is unavailable, leave it blank. Ensure the output is properly formatted as a table. Here's an example of the required format:

Qualification	Stream	CGPA	Passing Year
Bachelor's	Computer Science	8.5	2022
Master's	Data Science	9.0	2024

Now, analyze the following resume text and extract the details:
[Insert Resume Text Here]

RTCCF

[ROLE]

Act as a Chief Technology Officer (CTO).

[TASK]

Evaluate whether our company should build an AI-powered customer support assistant internally or purchase a commercial solution.

[CONTEXT]

Our company receives approximately 50,000 customer support requests every month. Current response times average 48 hours, and customer satisfaction has declined by 12%. Leadership wants to improve customer experience using AI.

[CONSTRAINTS]

- Budget limited to \$300,000
- Implementation within 6 months
- Must comply with data privacy regulations

[OUTPUT FORMAT]

Present the results in a comparison table containing:

- Option
- Estimated Cost
- Benefits
- Risks
- Time to Implement
- Recommendation

Provide an executive summary at the end.

Prompt Examples

Instruct the model to answer using a reference text [RAG]

- You will be provided with a document delimited by triple quotes and a question.
- Your task is to answer the question using only the provided document and to cite the passage(s) of the document used to answer the question.
- If the document does not contain the information needed to answer this question then simply write: "Insufficient information." If an answer to the question is provided, it must be annotated with a citation. Use the following format for to cite relevant passages ({"citation": ...}).
- `"""<insert document here>"""`
- Question: `<insert question here>`

Prompt Examples

RAG for Document

- Please find the attached document and a question.
- Your task is to answer the question using only the provided document and to cite the passage(s) of the document used to answer the question.
- If the document does not contain the information needed to answer this question then simply write: "Insufficient information." If an answer to the question is provided, it must be annotated with a citation. Use the following format for to cite relevant passages (`{"citation": ..., "page": page_number, "paragraph": paragraph_number}`).
- Question: <insert question here>

Prompt Examples

META Prompting

Meta prompting is a technique where a prompt instructs an AI model on *how to think, structure, or generate its response*, rather than directly asking for the final answer.

In simple terms, it is “a prompt about prompting”

I have created a project in chatgpt for HR policy document, I want that when the user asks question, the system provide the response I have created a prompt, please enhance this, add a suitable role play as well. Please find the attached document and a question. Your task is to answer the question using only the provided document and to cite the passage(s) of the document used to answer the question. If the document does not contain the information needed to answer this question then simply write: "Insufficient information." If an answer to the question is provided, it must be annotated with a citation. Use the following format for to cite relevant passages ({"citation": ..., "page": page_number, "paragraph": paragraph_number}).

Prompt Examples

Chain of thoughts

Chain-of-thought prompting encourages the model to reason step-by-step to arrive at the final answer, improving performance on tasks that require logical reasoning or complex calculations.

Use the following step-by-step instructions to respond to user inputs.

Step 1 - The user will provide you with text in triple quotes. Summarize this text in one sentence with a prefix that says "Summary: ".

Step 2 - Translate the summary from Step 1 into French, with a prefix that says "Translation: ".

""insert text here""

Prompt Examples

Chain of thoughts

Chain-of-thought prompting encourages the model to reason step-by-step to arrive at the final answer, improving performance on tasks that require logical reasoning or complex calculations.

Use the following step-by-step instructions to respond to user inputs.

Step 1 - The user will provide you with text in triple quotes. Summarize this text in one sentence with a prefix that says "Summary: ".

Step 2 - Translate the summary from Step 1 into Spanish, with a prefix that says "Translation: ".

"""" **Generative artificial intelligence (generative AI, GenAI,^[1] or GAI)** is a subset of artificial intelligence that uses generative models to produce text, images, videos, or other forms of data.^{[2][3][4]} These models learn the underlying patterns and structures of their training data and use them to produce new data^{[5][6]} based on the input, which often comes in the form of natural language prompts.^{[7][8]}

Improvements in transformer-based deep neural networks, particularly large language models (LLMs), enabled an AI boom of generative AI systems in the 2020s. These include chatbots such as ChatGPT, Copilot, Gemini, and LLaMA; text-to-image artificial intelligence image generation systems such as Stable Diffusion, Midjourney, and DALL-E; and text-to-video AI generators such as Sora.^{[9][10][11][12]} Companies such as OpenAI, Anthropic, Microsoft, Google, and Baidu as well as numerous smaller firms have developed generative AI models.^{[7][13][14]}

Generative AI has uses across a wide range of industries, including software development, healthcare, finance, entertainment, customer service,^[15] sales and marketing,^[16] art, writing,^[17] fashion,^[18] and product design.^[19] However, concerns have been raised about the potential misuse of generative AI such as cybercrime, the use of fake news or deepfakes to deceive or manipulate people, and the mass replacement of human jobs.^{[20][21]} Intellectual property law concerns also exist around generative models that are trained on and emulate copyrighted works of art.^[2]

""""

Chain of thoughts

Math Reasoning:

A store is offering a 20% discount on a jacket that originally costs \$80. After applying the discount, a 5% sales tax is added. What is the final price? Think step by step.

Business Decision-Making:

A company is deciding between launching a new product in Q3 or Q4. Q3 has higher demand but limited production capacity, while Q4 has more competition. What should the company do? Explain your reasoning step by step.

HR Dilemma:

An employee is consistently underperforming but recently disclosed a personal health issue. As their manager, how should you proceed? Think through the steps involved in a fair and compassionate approach.

Supply Chain Optimization:

You have three suppliers: one is fast but expensive, one is cheap but slow, and one has inconsistent quality. Your goal is to ensure on-time delivery and control costs. How do you decide which supplier(s) to use? Reason it out step by step.

Marketing Campaign Strategy:

You need to create a campaign for a new eco-friendly product. The budget is limited, and the target audience is Gen Z. How would you design this campaign? Think step by step from audience insight to platform choice to message.

Prompt Examples

One Shot Learning

In one-shot learning, the model is given a single example within the prompt to learn the task. Here's an example using a task of converting a sentence into the passive voice.

Prompt: "You are a grammar expert.

Convert sentences from active voice to passive voice.

Example:

Active: The chef cooked a delicious meal.

Passive: A delicious meal was cooked by the chef.

Now, convert the following sentence:

Active: The cat chased the mouse."

One Shot Learning

Text Classification (Sentiment Analysis):

Text: "The customer service was amazing and the staff was super helpful."

Sentiment: Positive

Now, classify this:

Text: "I had to wait for 30 minutes and my issue is still unresolved."

Sentiment:

Named Entity Recognition (NER):

Text: "Elon Musk founded SpaceX in 2002."

Entities: [Elon Musk: PERSON], [SpaceX: ORGANIZATION], [2002: DATE]

Now, extract entities from this:

Text: "Sundar Pichai became CEO of Google in 2015."

Entities:

Email to Action Item Extraction:

Email: "Please prepare the Q2 performance report and share it by Monday."

Action Item: Prepare Q2 performance report by Monday

Now, extract the action item from this email:

Email: "Can you book the conference room for Friday's team meeting?"

Action Item:

Prompt Examples

Few Shot Learning

In few-shot learning, the model is provided with several examples within the prompt to learn the task before attempting a new instance.

Task: Sentiment Analysis - Identify whether a given review is Positive or Negative.

Prompt: "You are a sentiment analysis expert. Determine whether the sentiment of each review is Positive or Negative."

Examples:

- Review: The movie was fantastic and kept me on the edge of my seat!
 - Sentiment: Positive
- Review: The food was cold, and the service was terrible.
 - Sentiment: Negative
- Review: I absolutely loved the book; it was a page-turner!
 - Sentiment: Positive

Now analyze the following review:

- Review: The product quality was poor, and it broke after one use."

Few Shot Learning

Intent Classification (Customer Support Domain):

Example 1:

Text: "I forgot my password and can't log in."

Intent: Account Recovery

Example 2:

Text: "I want to cancel my subscription."

Intent: Cancellation Request

Example 3:

Text: "How do I change my billing address?"

Intent: Update Billing Info

Now classify the intent:

Text: "Can you help me switch to a yearly plan?"

Intent:

Few Shot Learning

Product Categorization (Retail Domain):

Example 1:

Product: "Nike running shoes for men"

Category: Footwear > Men's Shoes > Running Shoes

Example 2:

Product: "Samsung 55-inch Smart TV"

Category: Electronics > Televisions > Smart TVs

Example 3:

Product: "Wooden dining table with 6 chairs"

Category: Furniture > Dining Room > Dining Sets

Now categorize this product:

Product: "Apple AirPods Pro with noise cancellation"

Category:

Prompt Examples

Multi-Turn Prompting

Guide the model step-by-step, asking clarifying questions or refining answers.

You are an experienced technical interviewer conducting a mock interview for a software developer position. Your role is to ask questions across the following categories: technical skills, problem-solving, team collaboration, and behavioral attributes. Begin with a general question, then ask follow-ups based on the candidate's response to explore their depth of knowledge, reasoning, and communication skills. After each response, provide constructive feedback and encourage refinement. Conclude the interview with a brief evaluation of their performance.

Ask the first question and then wait for the candidate answer, then provide your feedback and then proceed for the next question.

LLM as Judge

Description: The model is prompted to act as an impartial evaluator, scoring or comparing responses based on specific criteria like clarity, logic, or helpfulness.

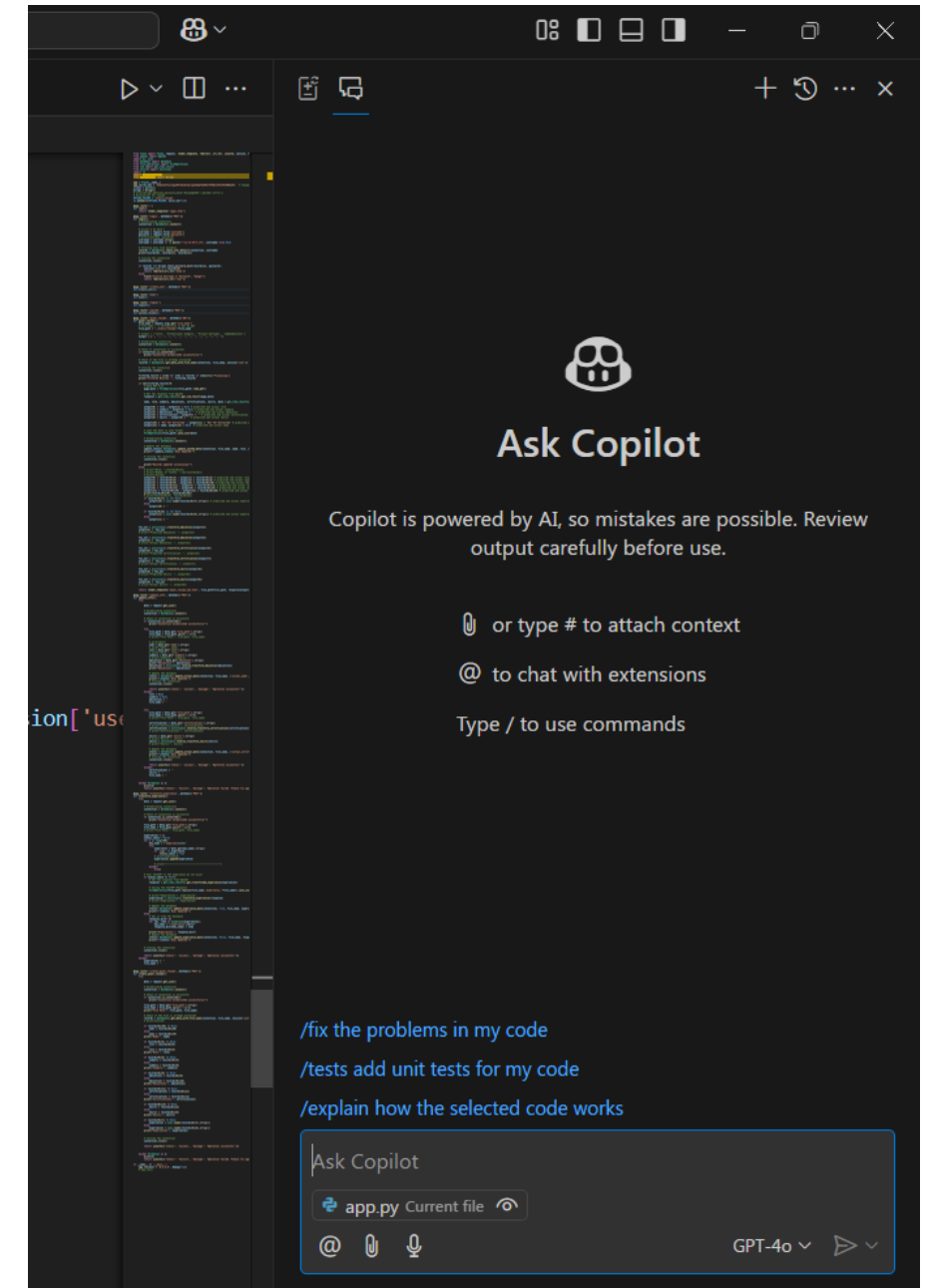
Prompt Example:

"Two teams proposed ad campaigns for a new kitchen appliance. Judge which campaign is more likely to succeed based on clarity, appeal, and market reach. Justify your verdict."

Prompt Examples

Coding Together [VS Code + Copilot]

```
class Solution {  
    public ListNode addTwoNumbers(ListNode l1, ListNode l2) {  
        ListNode head1 = l1;  
        ListNode temp1 = head1;  
  
        ListNode head2 = l2;  
        ListNode temp2 = head2;  
  
        ListNode result = null;  
  
        while (temp1 != null && temp2 != null) {  
            int sum = temp1.val + temp2.val;  
            if (result == null) {  
                result = new ListNode(sum);  
            } else {  
                result.next = new ListNode(sum);  
            }  
            temp1 = temp1.next;  
            temp2 = temp2.next;  
        }  
    }  
}
```



References

1. [Prompt Engineering](#)
2. [Prompt Engineering Guide for GPT-4](#)
3. [GPT-5 Prompt Engineering Guide](#)
4. [OpenAI Cookbook](#)

**Thanks for
your time**